

# Build THINKING MACHINES With OpenCog

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The human mind is arguably the most complex in this world. Utterly fascinating, it is an object of admiration to everyone. Some people take this admiration to a higher level and try to understand how the brain does what it does. Others go even further and attempt to replicate this marvel of a creation. While this is no mean task, science is a lot closer to creating artificial intelligence (AI) today than it was a few years ago.

This article takes you through a tool that provides an environment for R&D in AI domain. Be it to make a software application smarter or to create your own clone, this tool provides the very basic raw materials you need to start with, and may be carves out a path and guides you along. But, it is no cake-walk. The tool we are talking about is OpenCog.

## Apt for research in AI

OpenCog is an open source software project, which aims to create an open source framework for artificial general intelligence. The developers state, "We are undertaking serious effort to build a thinking machine." While this is stretching it too far for the present, the team behind OpenCog presently aims at nurturing the R&D environment and creating a code that can help make smarter systems, and working towards de-

veloping cognitive capability at the human level.

OpenCog has been developed by OpenCog Foundation under GNU Affero general-public licence and is supported by Artificial General Intel-

## Keeping the first foot forward

Prime inspiration for the project is the artificial general intelligence theories by Dr Ben Goertze. The concept of probabilistic logic net-

works is made extensive use of, to robustly handle uncertainty and perform effective reasoning in real-world circumstances. The idea is a novel conceptual, mathematical and computational approach to uncertain inference.

Well, you have seen what to expect from this tool, but it is still not clear as to how you can go about working with it. Building a cognitive system is easier said than done, and certainly that much more complicated for someone just beginning to work with AI. Let us begin by trying to understand how OpenCog goes about creating AI.

**The knowledge base.** The first step is to create a system that stands for the knowledge inside the memory of an intelligent system, called knowledge representation. While this details the basic nature and organisation of knowledge components, it could be highly complex and dynamic, or simple, static and easy to describe. The catch here is how explanatory you want the knowledge representation to be. Does it have to be a repeated looping of a basic element, specialis-

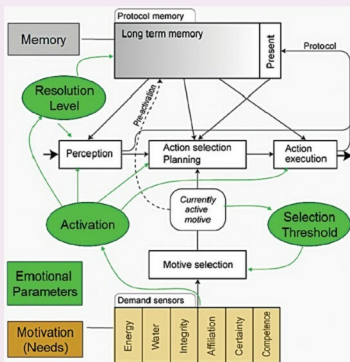


Fig. 1: MicroPsi architecture, an implementation of OpenPsi

ligence Research Institute (AGIRI), Google Summer of Code project and others. It is written using a combination of C++, Python and Scheme, and is built for Linux platform. The software is currently under an early stage of development, and is mostly useful for building applications that require significant AI capability. This is an interesting and powerful AI platform, but you will need to make significant customisations and integrations to get your projects up and running.